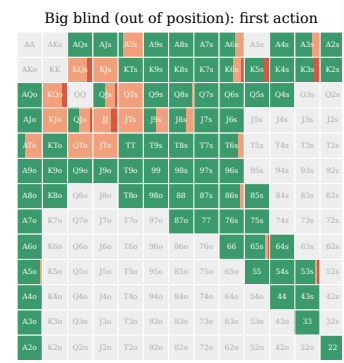
Solving Poker

The numerical analysis behind poker solvers
— and a second-order finish for them

Numerical Analysis Research Club

POKER Ten years of superhuman poker

- **2015** Cepheus: heads-up *limit* hold'em "essentially weakly solved" with CFR$^+$
  Bowling, Burch, Johanson, Tammelin, *Science*
- **2017** DeepStack beats 33 professionals at heads-up *no-limit* (44,852 hands, ≈490 mbb/hand)
  Moravčík et al., *Science*
- **2017** Libratus beats four top pros over 120,000 hands (14.7 big blinds / 100 hands)
  Brown & Sandholm, *Science* 2018
- **2019** Pluribus: six-player no-limit
  Brown & Sandholm, *Science*
- **Today**: every serious player studies with solvers.



A solver's answer (we computed): how the big blind should play every hand on one river.

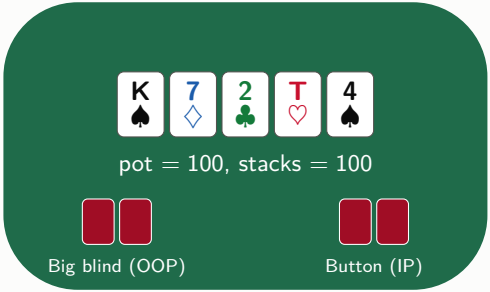
The plan: poker $\leftrightarrow$ math



- The equilibrium of a poker game is the solution of a (huge, structured) bilinear saddle-point problem.
- Solvers use a *first-order*, matrix-free iteration: cheap steps, slow tail.
- Our contribution: treat the end of the solve as a *nonsmooth equation* on the solver's predicted support and finish with a proximal semismooth Newton method — machine precision where CFR needs millions of iterations.

POKER Texas hold'em in one slide

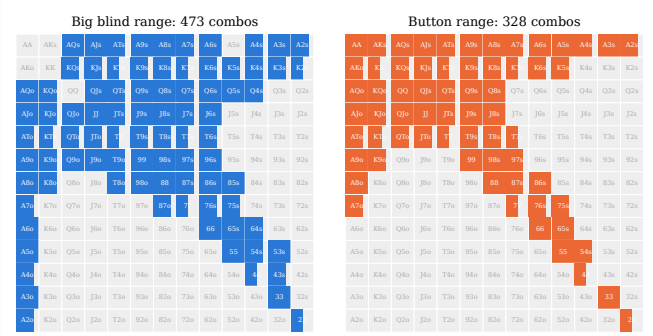
- Two players, each dealt two private cards.
- Five public *board* cards revealed in rounds: flop (3), turn (1), river (1).
- A betting round before and after each reveal: *check, bet, call, raise, fold*.
- *No-limit*: any bet size up to your stack.
- Best five-card hand from seven cards wins at showdown — unless everyone else folded.



Our running example: the river, last betting round.

The big blind acts first ("out of position").

POKER Ranges: what a solver takes as input



- A **range** = a probability vector over two-card combinations: $\binom{52}{2} = 1326$, and $\binom{47}{2} = 1081$ once the board is known.
- Here: big blind called a raise, Button opened (illustrative).
- Bar width = combinations still possible given the board: *card removal* (K♠ on board ⇒ fewer Kx).
- Output: for every public node and every hand, a distribution over actions.

Two-player zero-sum games and Nash equilibrium

Strategies $x \in X$, $y \in Y$ (compact convex), payoff to player 1: bilinear $u(x, y) = x^\top A y$.

Minimax theorem (von Neumann 1928)

$$\max_{x \in X} \min_{y \in Y} x^\top A y \;=\; \min_{y \in Y} \max_{x \in X} x^\top A y \;=:\; v^* \quad \text{(the value of the game)}.$$

A pair (x^*, y^*) attaining both sides is a **Nash equilibrium**.

The error metric: exploitability

$$\mathrm{expl}(x, y) \;=\; \tfrac{1}{2}\Big(\max_{x'} x'^\top A y \;-\; \min_{y'} x^\top A y' \Big) \;\geq 0, \qquad \mathrm{expl}(x, y) = 0 \iff \text{equilibrium}.$$

- Playing x^* guarantees $\geq v^*$ against *anyone*: unexploitable, "game-theory optimal".
- expl is a duality gap: the residual every solver reports. Units: chips per hand.

POKER Why equilibria mix: bluffing and pot odds

Toy river: pot 1, bettor has either the nuts or nothing; bets s; caller holds a bluff-catcher.

- Caller indifferent $\Rightarrow$ bettor's bluffs are a fraction

$$\beta = \frac{s}{1+2s} \quad \text{of the betting range.}$$

- Bettor's bluffs indifferent $\Rightarrow$ caller defends with frequency

$$\text{MDF} = \frac{1}{1+s} \quad (\text{"minimum defence frequency"}).$$

- Pure strategies are exploitable; equilibrium *randomises* exactly at indifference.

bet s	bluffs β	MDF
1/3	20%	75%
3/4	30%	57%
1	33%	50%
2	40%	33%

Indifference conditions = complementary slackness.

(We will see the solver reproduce the MDF column.)

MATH Kuhn poker: the smallest interesting example

- Deck  ; each player antes 1, gets one card.
- One bet of size 1; actions check/bet, then fold/call.
- Equilibria form a *one-parameter family* (Kuhn 1950): first player
  - bets J with prob. $\alpha \in [0, \frac{1}{3}]$ (a bluff),
  - bets K with prob. 3α (slow-plays the rest),
  - value $v^* = -\frac{1}{18}$.

We computed (Kuhn)

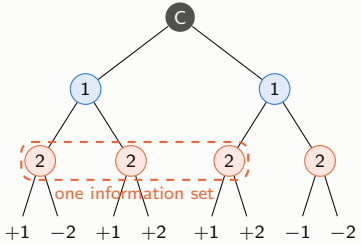
prox-Newton output: $\alpha = 0.198$, K bets $0.593 = 3\alpha$
value $-0.0555\ldots = -1/18$
exploitability 1.1×10^{-16}

The equilibrium set is a **polytope**, not a point: any method must cope with *non-isolated* solutions.

Extensive-form games

- Game tree: chance nodes (deals), decision nodes, terminal payoffs.
- **Information set** I: histories a player cannot distinguish (same public actions, opponent's cards unknown).
- Behavioural strategy: a distribution $\sigma(I) \in \Delta(A(I))$ at every information set.
- Perfect recall: you never forget what you knew or did.

game	information sets
Kuhn	12
Leduc hold'em	288
HU limit hold'em	3.2×10^{14}
HU no-limit hold'em	$\sim 10^{161}$



Player 2 sees the action, not the deal.

MATH The sequence form: equilibrium is a linear program

Sequences of player 1: paths of her own actions; *realisation plan* $x_\sigma = \prod$ of her action probabilities along σ.

$$x_\emptyset = 1, \qquad \sum_{a \in A(I)} x_{\sigma_I a} = x_{\sigma_I} \ \ \forall I, \qquad x \geq 0 \qquad \iff \qquad Ex = e,\ x \geq 0 \quad \text{(a "treeplex")}$$

Payoff is *bilinear* in realisation plans: $u(x, y) = x^\top A y$,
$A_{\sigma\tau} = \sum_{\text{leaves } z \text{ reached by } (\sigma,\tau)} \text{chance}(z)\,\text{payoff}(z)$.

Koller–Megiddo–von Stengel (1994/96), von Stengel (1996)

$$\max_{x,\,q} \ f^\top q \quad \text{s.t.} \quad A^\top x - F^\top q \geq 0, \quad Ex = e, \quad x \geq 0$$

— an LP whose size is *linear* in the game tree (the normal form is exponential).

- Dual variables q_J = values of the opponent's information sets J (= counterfactual best-response values).
- So: why not just call an LP solver?

POKER Where the LP breaks: the river spot

Ranges (illustrative single-raised pot):

Big blind: 473 hand combinations; Button: 328.

Bets: 33%, 75%, all-in; raises: all-in.

Sequence form (we computed):

- 7,569 + 5,249 sequences, 4,808 information-set rows;
- A has 2.9 million nonzeros: every leaf contributes a *dense* hand×hand block.

We computed (one core)

HiGHS LP: not finished after 25 min

DCFR, 5,000 iterations: exploitability 1.3×10^{-4} chips in 16 s

The payoff blocks are what LP and sparse direct methods choke on — and what CFR handles with a few matrix–vector products.

MATH Regret matching (Hart & Mas-Colell 2000)

Repeated decision, actions A, payoffs $u_t(a)$ revealed each round. No step size, no tuning:

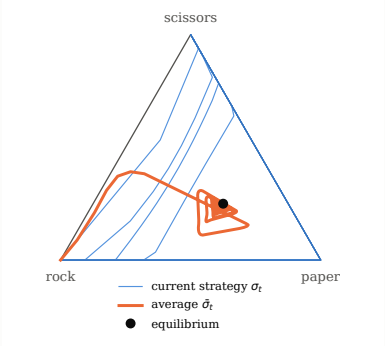
$$R_T(a) = \sum_{t=1}^{T} \big(u_t(a) - u_t(\sigma_t)\big), \quad \sigma_{T+1}(a) = \frac{[R_T(a)]^+}{\sum_b [R_T(b)]^+}.$$

Blackwell approachability ⇒ no regret

$$\max_a R_T(a) \;\le\; \Delta\sqrt{|A|\,T} \qquad (\Delta = \text{payoff range}).$$

Self-play ⇒ equilibrium

Average regrets $\le \varepsilon$ for both players ⇒ the *average* strategies are a 2ε-equilibrium.



We computed: self-play on a weighted rock–paper–scissors. The iterate cycles; the average converges to $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$.

MATH Counterfactual regret minimisation (Zinkevich et al. 2007)

Run regret matching *locally at every information set*, on **counterfactual values**

$$v_i(\sigma, I, a) = \sum_{h \in I} \pi^{\sigma}_{-i}(h) \sum_{z \sqsupset ha} \pi^{\sigma}(ha \to z)\, u_i(z)$$

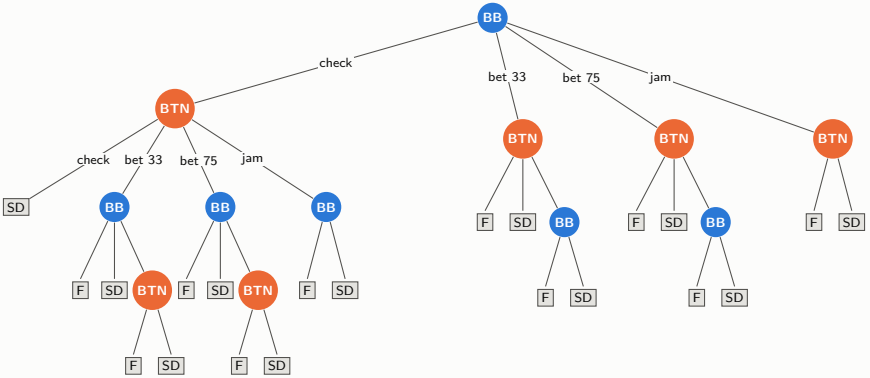
(opponent and chance reach probabilities only — "as if I had tried to reach I").

Regret decomposition

$$R_T^{\text{full}} \;\le\; \sum_I \left[R_T^{\text{imm}}(I)\right]^+ \quad\Longrightarrow\quad \text{expl}(\bar{x}_T, \bar{y}_T) \;\le\; \frac{\Delta\, |\mathcal{I}| \sqrt{|A|}}{\sqrt{T}}.$$

- One iteration = one pass over the tree: compute reaches top-down, values bottom-up, update regrets.
- Memory: two numbers per (information set, action). Matrix-free: A is never formed.

The public tree of our river spot



12 decision nodes, 21 leaves (F = fold, SD = showdown). At each node the solver stores *vectors* over hands: reach $\pi_{\text{BB}} \in \mathbb{R}^{473}$, $\pi_{\text{BTN}} \in \mathbb{R}^{328}$, regrets and average strategy as $473 \times |A|$ or $328 \times |A|$ arrays. One CFR iteration = one sweep down (reaches) and up (values) this tree.

Vectorised CFR: one pass for all hands

At a public node, store vectors over private hands: reach $\pi_1 \in \mathbb{R}^{n_1}$, $\pi_2 \in \mathbb{R}^{n_2}$. At a leaf:

$$v_1 = \text{amount} \cdot B\,\pi_2, \qquad v_2 = -\text{amount} \cdot B^\top \pi_1, \qquad B_{ij} = \underbrace{C_{ij}}_{\text{no shared card}} \cdot \begin{cases} 1 & \text{fold} \\ \operatorname{sign}(s_i - s_j) & \text{showdown} \end{cases}$$

Structure (s = hand strength, sorted):

- $C = \mathbf{1}\mathbf{1}^\top - K$, K sparse (hands sharing a card: $\approx 8\%$ of pairs);
- $\operatorname{sign}(s_i - s_j)$ = lower-ones minus upper-ones: *semiseparable* (rank-1 off-diagonal blocks).

Consequence

$B\pi$ in $O(n \log n)$ by sorting + prefix sums + a per-card correction, instead of $O(n^2)$ (Johanson et al. 2011).

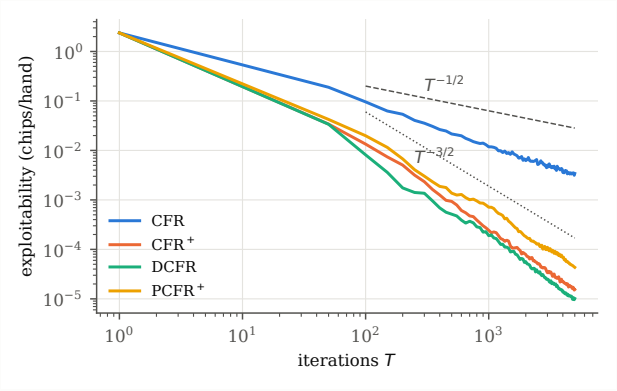
Full river: $n = 1081$ combinations per player $\Rightarrow$ the dense blocks behind the 2.9 M nonzeros.

MATH Modern CFR = a bag of numerical tricks

method	update	numerical reading
CFR$^+$ Tammelin 2014	$R \leftarrow [R + r]^+$; average weighted by t	projection onto the orthant; Cesàro $\rightarrow$ linear averaging
DCFR Brown–Sandholm 2019	positive regrets $\times \frac{t^{1.5}}{t^{1.5}+1}$, negative $\times \frac{1}{2}$; average $\times (\frac{t}{t+1})^2$	forgetting factors: down-weight the transient
PCFR$^+$ Farina et al. 2021	$\sigma_t \propto [R_{t-1} + m_t]^+$, prediction $m_t = r_{t-1}$	optimism / extragradient
alternating	update player 2 using player 1's *new* strategy	Gauss–Seidel vs Jacobi

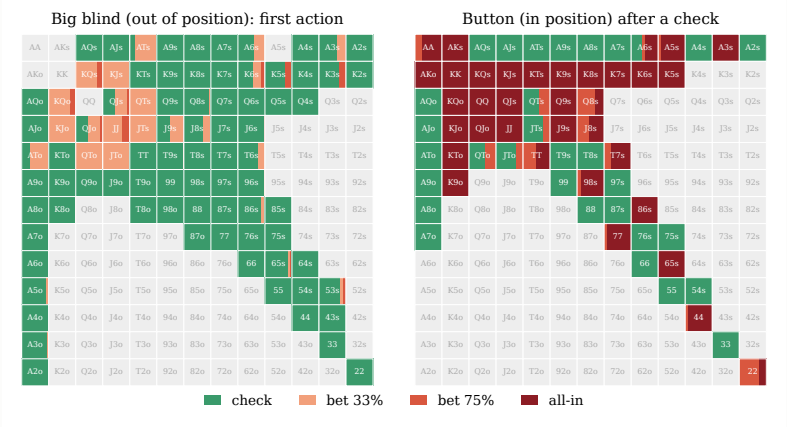
- All are matrix-free and parameter-light; worst-case theory is still $O(T^{-1/2})$.
- In practice: $\approx T^{-1}$ to $T^{-3/2}$ for averages; the *last iterate* oscillates.

MATH We computed: the baselines on Leduc hold'em



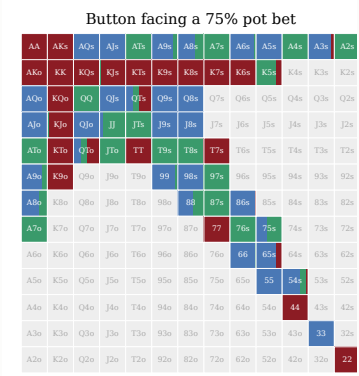
- Our range-vectorised implementation (alternating updates), exploitability of the *average* strategy.
- 5,000 iterations, 14 s: DCFR reaches 1.0×10^{-5}.
- 50,000 iterations, 130 s: CFR$^+$ 9.2×10^{-7}, DCFR 1.6×10^{-6}.
- Each extra digit costs $\approx 10\times$ more iterations.

POKER What a solution looks like: first actions



We computed (DCFR, 5,000 iterations, exploitability 1.3×10^{-4} chips). Upper triangle: suited; lower: offsuit; grey: not in range. Each cell shows the action mix, weighted over its combinations.

POKER Facing a bet: the solver rediscovers pot odds



defender	line	s	defends	MDF
Button	BB checks, BTN bets 75, BB jams	0.10	64.7%	90.9%
Button	BB bets 33	0.33	74.3%	75.2%
Button	BB checks, BTN bets 33, BB jams	0.40	57.8%	71.2%
Button	BB bets 75	0.75	55.8%	57.1%
Button	BB bets 100	1.00	49.1%	50.0%
Big blind	BB bets 75, BTN jams	0.10	63.4%	90.9%
Big blind	BB checks, BTN bets 33	0.33	52.2%	75.2%
Big blind	BB bets 33, BTN jams	0.40	56.3%	71.2%
Big blind	BB checks, BTN bets 75	0.75	37.5%	57.1%
Big blind	BB checks, BTN bets 100	1.00	32.6%	50.0%

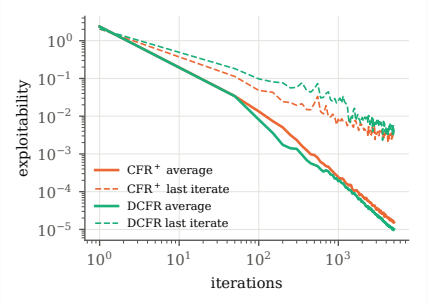
Button: MDF to within 1.5 points. Big blind *under*-defends: the Button's marginal bluffs have showdown value, so they are indifferent at EV>0, not at 0.

MATH Why push precision further?

Practical solvers stop at ≈ 0.1–0.5% of the pot.

Reasons to want more:

- **Subgame re-solving** stitches many solves together; errors compound across streets.
- **Exact mixing frequencies and supports**: which hands *exactly* mix?
- **Certificates**: a primal–dual pair with residual 10^{-13} proves near-optimality.
- **Research**: benchmarks need ground truth.



We computed (Leduc): only the average converges; the last iterate stalls.

The numerical analyst's view

CFR is a first-order method on a nonsmooth, degenerate saddle-point problem: robust, cheap per step, *sublinear* tail. The classical cure for a slow tail: a *local, higher-order* method started from a good point.

NEW Attempt 1 (didn't work): Anderson acceleration of CFR$^+$

- One CFR$^+$ sweep is a map G on the regret vector: $R \mapsto [R + r(\sigma(R))]^+$.
- Fixed points of $G \Leftrightarrow \sigma(R)$ is an equilibrium. Natural idea: accelerate the fixed-point iteration.
- Type-II Anderson acceleration (Anderson 1965; Walker–Ni 2011), memory 10, with safeguards (clip to ≥ 0, restart on residual growth).

We computed

Kuhn: exact equilibrium. Leduc: 2.3×10^{-4} after 2,000 map evaluations — DCFR has 1.9×10^{-4} after 1,000. No gain.

Why: G is piecewise linear and nonsmooth; its fixed points are non-isolated (whole rays, since $\sigma(cR) = \sigma(R)$); AA theory (Toth–Kelley 2015) needs a smooth contraction. We also tried vector extrapolation (RRE) of the averages, and Levenberg–Marquardt on the KKT system: same story.

NEW Reformulate: a monotone variational inequality

Unknowns $z = (x, y, p, q)$: both realisation plans and both dual (value) vectors. Equilibrium $\iff$ LP optimality for both players:

$$0 \le x \;\perp\; E^\top p - Ay \;\ge 0, \qquad Ex = e,$$
$$0 \le y \;\perp\; A^\top x - F^\top q \;\ge 0, \qquad Fy = f.$$

Equivalently, with $w(z) = Mz + b$:

$$M = \begin{pmatrix} 0 & -A & E^\top & 0 \\ A^\top & 0 & 0 & -F^\top \\ -E & 0 & 0 & 0 \\ 0 & F & 0 & 0 \end{pmatrix}, \qquad M + M^\top = 0.$$

Key facts

- M is **skew**: the VI is monotone, the solution set is a polyhedron (non-isolated, cf. Kuhn).
- p, q = information-set values; CFR's best-response values give them for free: $p_I \approx r(h)\,\mathrm{BR}(I, h)/Z$.

NEW Outer loop: proximal point (Rockafellar 1976)

$$z^{k+1} \;=\; \text{solution of the VI with } \; w_\rho(z) = (M + \rho I)\, z + b - \rho\, z^k.$$

- $M + \rho I$ is positive definite ($d^\top (M + \rho I) d = \rho \|d\|^2$) $\Rightarrow$ **unique** solution of each subproblem.
- Converges to a solution of the original VI for any $\rho > 0$.
- Polyhedral problem $\Rightarrow$ **linear** rate, factor $\to 0$ as $\rho \to 0$ (Luque 1984).
- Small ρ: fast outer loop, but ill-conditioned subproblems ($\|(M + \rho I)^{-1}\| \le 1/\rho$).

We computed (Leduc, $\rho = 10^{-5}$)

VI residual after each prox step:

$3.4 \times 10^{-2} \to 4.8 \times 10^{-6} \to 3.3 \times 10^{-9}$
$\to 6.9 \times 10^{-12} \to 2.0 \times 10^{-14}$

$\approx$ three digits per step.

NEW Inner loop: semismooth Newton = primal–dual active set

Each subproblem is a piecewise-linear equation:

$$\Psi(z) = \Big(\min\big(x_i,\ w_\rho(z)_i\big) \Big)_{\text{bounded}}, \quad \big(w_\rho(z)_j\big)_{\text{free}} = 0.$$

A generalised Jacobian row is $e_i^\top$ ("$x_i = 0$", active) or a row of $M + \rho I$ ("indifferent", inactive): $V = D_a + D_b(M + \rho I)$. The Newton step = fix the active set, solve one linear system (Hintermüller–Ito–Kunisch 2002).

Lemma. Every $V = D_a + D_b(M + \rho I)$, $D_a, D_b \geq 0$ diagonal, $D_a + D_b \succ 0$, is nonsingular.

If $Vd = 0$, put $u = (M + \rho I)d$. Row i: $(D_a)_{ii} d_i + (D_b)_{ii} u_i = 0$ with nonnegative, not both zero coefficients $\Rightarrow d_i u_i \leq 0$. Hence $\rho \|d\|^2 = d^\top u = \sum_i d_i u_i \leq 0$, so $d = 0$. □

- Right active set $\Rightarrow$ **exact** solution in one step (finite termination). Wrong one $\Rightarrow$ no global guarantee.
- The lemma also covers Fischer–Burmeister Newton (globalised by a line search), which we tried first: robust, but on larger spots the line search crawls (steps $\sim 10^{-3}$).

NEW Globalisation: interior point, then crossover

First prox subproblem (most active-set changes):

- Mehrotra predictor–corrector **interior-point** method, started from the CFR point; its iteration count does not care how many actions enter or leave the support.
- Its point predicts the active set $(x_i > s_i) \Rightarrow$ active-set Newton steps land *exactly* on the subproblem's solution: **crossover**, as in LP solvers.

Later subproblems: active-set Newton from the previous solution (successive prox points share their active set): 1–3 linear solves each.

One exact solve

1. warm start: CFR average $(\bar{x}, \bar{y})$, duals (p, q) from one best-response pass;
2. prox step 1: IPM + crossover;
3. prox steps $2, 3, \ldots$: active-set Newton;
4. stop at VI residual 10^{-12}.

$\rho = 10^{-5}$ throughout.

NEW The linear algebra: a quasi-definite system

Every Newton-type system is $(D + M + \rho I)\, d = r$, $D \geq 0$ diagonal. Flip the signs of the y- and p-rows:

$$K = \begin{pmatrix} D_x + \rho I & -A & E^\top & 0 \\ -A^\top & -(D_y + \rho I) & 0 & F^\top \\ E & 0 & -\rho I & 0 \\ 0 & F & 0 & \rho I \end{pmatrix} \quad \text{symmetric.}$$

- Grouping (x, q), (y, p): $\left(\begin{smallmatrix} P & B \\ B^\top & -N \end{smallmatrix}\right)$, $P, N \succ 0$: **quasi-definite**.
- Vanderbei (1995): $LDL^\top$ for *every* symmetric ordering $\Rightarrow$ AMD, no pivoting.
- IPM systems: $LDL^\top$ + 2 refinements. Active-set systems ($\pm\rho$ diagonal): pivoted LU on the support.

We computed: one factorisation

	small river	medium river
unsymmetric LU	0.11 s	25 s
quasi-def. $LDL^\top$	0.04 s	7.6 s

SuperLU (COLAMD) vs QDLDL (AMD), one core.

POKER NEW Poker makes the Newton step cheaper

- Active-set steps solve only on the *support* + information sets: dropped actions are eliminated.
- Equilibria in poker are *mostly pure*: mixing happens only at indifference.
- So the expensive dense hand×hand blocks of A shrink to the hands that actually reach each leaf.

We computed (exact equilibria)

	Leduc	small river
pure (infoset, hand)	49%	71%
sequences in support	59%	52%
full system size	964	2,864

POKER NEW The full method: restrict, solve exactly, verify

1. **DCFR**, a few thousand iterations (cheap, global).
2. **Predict the support**: keep actions with average probability $\geq \tau = 10^{-3}$.
3. **Restrict**: keep only information sets that *both* players can reach with kept actions (card removal included). Everything else stays at the CFR strategy.
4. **Solve the restricted VI exactly**: proximal point, IPM + crossover, active-set Newton.
5. **Verify** in the full game: a pruned action with positive counterfactual regret is a profitable deviation $\Rightarrow$ add it back, go to 3.

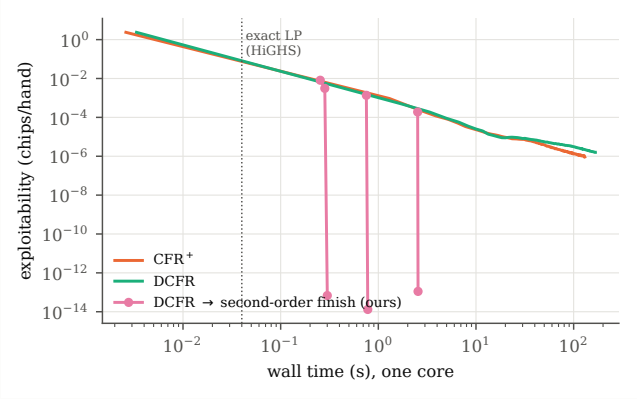
Why restrict?

The equilibrium set is *flat* in exactly the removed directions: never-played actions, and positions the opponent never lets you reach. They make the full VI ill-conditioned; the restricted one is not.

Certificate

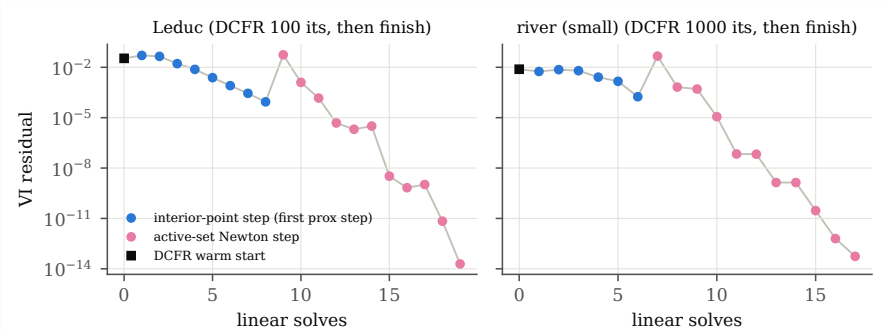
Step 5 computes exact best responses: the final exploitability is *measured* in the full game, not assumed.

NEW We computed: Leduc hold'em



- DCFR 100 iterations + finish (2 rounds, 26 linear solves): exploitability 6.8×10^{-14} in 0.30 s.
- CFR^{+} after 130 s: 9.2×10^{-7}.
- Exact LP (HiGHS): 0.04 s. At this size, the LP wins — the point is the *shape* of the curves.

NEW Inside one exact solve: every linear solve



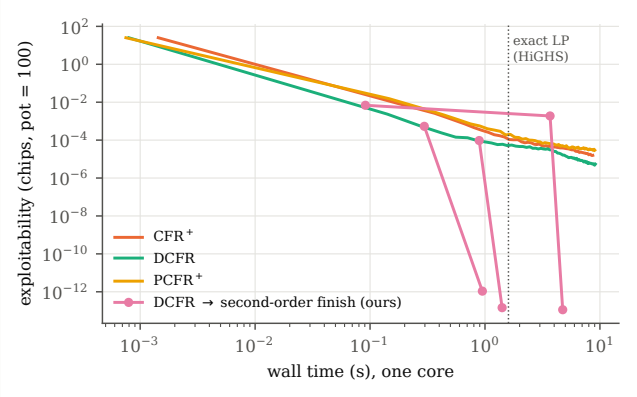
- Blue: interior-point iterations of the first prox subproblem (steady, insensitive to the active set).
- Pink: active-set Newton steps — crossover onto the exact subproblem solution, then 1–3 solves per prox step, three digits each.
- Total: 19 (Leduc) and 17 (small river) linear solves from the DCFR point to 10^{-13} (full, unrestricted VI shown).

POKER NEW Live demo: `python3 solver/demo.py` (1 second)

```
River Ks 7d 2c Th 4s: big blind 147 combos vs button 139 combos, pot 100, stacks 100
  DCFR     1 iterations   exploitability 2.51e+01 chips
  DCFR   250 iterations   exploitability 9.64e-03 chips
  DCFR   500 iterations   exploitability 2.35e-03 chips
  DCFR   750 iterations   exploitability 1.16e-03 chips
  DCFR  1000 iterations   exploitability 5.34e-04 chips
restrict to the predicted support, solve exactly, verify:
  round 0: 1896 unknowns, 16 linear solves, full-game exploitability 1.08e-12 chips, 0 actions added back
DCFR 0.3s, finish 0.6s;  game value for the big blind: +0.3122492839 chips

Big blind, first action  (check, bet0.75, allin):
  TcTd: check 100.0%  bet0.75   0.0%  allin   0.0%
  7h7c: check  68.4%  bet0.75   0.0%  allin  31.6%
  AhKd: check   0.0%  bet0.75 100.0%  allin   0.0%
  KhQh: check 100.0%  bet0.75   0.0%  allin   0.0%
  AsQs: check 100.0%  bet0.75   0.0%  allin   0.0%
  JcTc: check 100.0%  bet0.75   0.0%  allin   0.0%
  QsJs: check   0.0%  bet0.75  98.0%  allin   2.0%
```

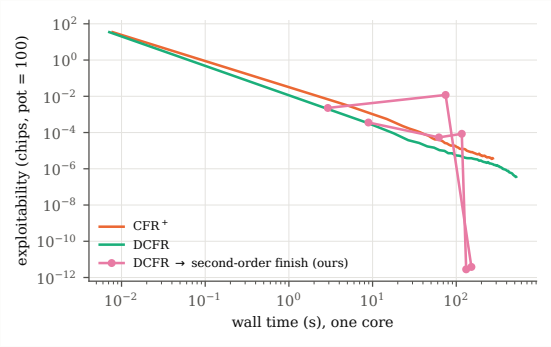
NEW We computed: a real river spot (small ranges)



Board K♠7♢2♣T♡4♠, 147 vs 139 combos, bet 75% / all-in.

- Ours: 1.1×10^{-12} chips in 0.95 s (DCFR 1,000 its, 1 round, 16 linear solves).
- DCFR: 5.4×10^{-6} after 30,000 iterations (9 s).
- HiGHS LP: 1.6 s.

NEW We computed: the medium spot (473 vs 328 combos, 3 bet sizes)



17,626 unknowns; A has 2.9 M nonzeros. Points = verification rounds.

- DCFR: 3.9×10^{-6} at 131 s; 3.6×10^{-7} after 190,000 iterations (520 s).
- Ours: DCFR 3,000 its, then 3 rounds on a problem with 34% of the unknowns (16 actions added back): 2.8×10^{-12} in 131 s.
- Without the restriction the finish *fails* here: 3.2×10^{-5} after 528 s. The full VI is flat ($\|(M+\rho I)^{-1}\| \approx 1/\rho$): an interior point with residual 4×10^{-6} still predicts a wrong active set.
- LP (HiGHS): not finished after 25 min.

NEW Where it stops winning: full ranges, and open problems

We computed: full 1,081-combo ranges for both players (47,568 unknowns, A: 22.4 M nonzeros)

Restricted problem 43% of the unknowns, one factorisation ≈10 s. Verification round 1 alone: 201 linear solves, 50 min, and 24 actions to add back. DCFR in 10 min: 9.5×10^{-6}. **Here CFR wins.**

- **Structured linear algebra.** Each payoff block is semiseparable + sparse (slide on vectorised CFR): products cost $O(n \log n)$, so Newton–Krylov steps would cost a few CFR iterations — if the shifted-skew systems $D + \rho I +$ skew get a good preconditioner (hierarchical / HSS factorisations of the blocks?).
- **Smarter support prediction** from CFR regrets ⇒ fewer verification rounds and smaller systems.
- **Inside a real-time agent**: re-solving subgames to high precision, where consistency between solves matters.

NEW What is new here, and what is not

- **Not new**: sequence-form LP (von Stengel 1996); proximal point (Rockafellar 1976); semismooth Newton / active-set methods (Qi–Sun 1993; Hintermüller–Ito–Kunisch 2002); interior point + crossover; quasi-definite $LDL^\top$ (Vanderbei 1995); the CFR family.
- **Closest work**: Yang, Gao, Lin & Kroer (arXiv 2512.12910, 2025): semismooth Newton for *matrix* games via Douglas–Rachford splitting, warm-started by PRM$^+$; experiments on normal-form games (Kuhn as a 27×64 matrix).
- **Here**:
  - a second-order finish directly on the *sequence-form* complementarity system of extensive-form poker;
  - skew-monotone VI + proximal regularisation $\Rightarrow$ uniformly nonsingular Newton matrices (one-line proof) and quasi-definite linear algebra;
  - CFR average + best-response values as a primal–dual warm start;
  - restriction to the CFR-predicted support (removes the flat directions) + verification by exact best responses;
  - tested on Leduc and no-limit river subgames with card removal.
- Early work solved small abstracted games exactly as LPs (Rhode Island hold'em; Gilpin & Sandholm 2005); to our knowledge no CFR-based solver hands over to a second-order method for the last digits.

Takeaways

1. A poker equilibrium is the solution of a structured bilinear saddle-point problem; the sequence form makes it an LP of size linear in the tree.
2. CFR = regret matching at every information set: a matrix-free, nonsmooth first-order method with a provable $O(T^{-1/2})$ rate and a much better practical one — plus numerical tricks (projection, discounting, optimism, Gauss–Seidel).
3. The solver's output is readable poker: polarised betting, block bets, pot-odds defence (to within 1.5%).
4. Read the equilibrium conditions as a **skew-monotone VI**: proximal point + semismooth (active-set) Newton, started by an interior-point crossover, turns CFR's slow tail into a few dozen linear solves.
5. The catch is **flatness** of the equilibrium set; restricting to CFR's predicted support and verifying afterwards fixes it: 10^{-12} in about two minutes on a 17,626-unknown river spot where DCFR is at 10^{-6}. With full ranges the linear algebra becomes the bottleneck and CFR wins again.

Code: `github.com/dwangnumbers/Poker-Solver` — solver, benchmarks, figures, these slides.

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Backup: implementation summary

`games.py`	Kuhn, Leduc, no-limit river builder (7-card evaluator, card removal, range notation)
`cfr.py`	range-vectorised public-tree CFR / CFR$^+$ / DCFR / PCFR$^+$, alternating updates, best response, exploitability
`seqform.py`	sequence form (E, F, A), dual estimates from best responses; `ProxIPMPDAS` (the method of the talk), FB variants, earlier attempts
`lp.py`	reference LP via HiGHS
`aa.py`, `extrap.py`	the negative results (Anderson, RRE)
`bench_*.py`, `fig_*.py`	every number and figure in these slides

- Exploitability: $\frac{1}{2}(\mathrm{BR}_1 + \mathrm{BR}_2)$ by exact best-response passes, normalised by the deal weight Z.
- Sequence form checked against the LP: Leduc value -0.0856064240780 (both).
- Dual warm start: $p_I = r_1(h)\,\mathrm{BR}_1(I,h)/Z$, $q_J = -r_2(h)\,\mathrm{BR}_2(J,h)/Z$.

Backup: the proximal-point rate, and why flat equilibria hurt

- $T(z) = Mz + b + N_{\mathcal{K}}(z)$ is maximal monotone; the step $z^{k+1} = (I + \rho^{-1}T)^{-1}z^k$ is a resolvent: firmly nonexpansive, Fejér monotone towards the solution set Z.
- If T^{-1} is (upper) Lipschitz at 0 with modulus a — true for polyhedral T (Robinson 1981) — then (Rockafellar 1976; Luque 1984)

$$\operatorname{dist}(z^{k+1}, Z) \;\le\; \frac{a\rho}{\sqrt{1 + a^2\rho^2}}\; \operatorname{dist}(z^k, Z).$$

- a measures how far a small residual can be from Z: it is large when the equilibrium set is "flat" (many hands whose actions barely matter).
- Fast outer loop needs $\rho \ll 1/a$; but $\|(M + \rho I)^{-1}\| \approx 1/\rho$ then makes each subproblem's active set fragile. Leduc and the small spot: observed factor $\approx 10^{-3}$ per step at $\rho = 10^{-5}$ (so $a\rho \approx 10^{-3}$); on the full medium spot no ρ we tried gave both — the support restriction removes the flat directions instead.

Backup: effect of the proximal parameter

We computed (Fischer–Burmeister variant; Leduc, warm start DCFR 200 iterations, final exploitability $\leq 10^{-13}$ in all cases):

ρ	10^{-3}	10^{-4}	10^{-5}	10^{-6}
prox steps	19	6	4	3

Continuation ($\rho_k \downarrow$) did not reduce the total number of Newton steps on the small river spot (54 fixed vs 63–96).